

J/ψ radiative corrections — status report

16–17 September 2026

1 The final formula is missing from the NLO paper

The NLO paper gives δ_{self} (8), δ_{vert} (12), δ_{soft} (13), δ_{hard} only as an unevaluated integral (14), and the exact Bardin–Shumeiko result (28). **There is no compact closed form for the total ultrarelativistic correction** — the quantity an experiment actually applies. The right one is Eq. (24)/(26) of Ehlotzky & Mitter with the missing term restored:

$$\delta(m_V, \omega_{cut}) = -\frac{4\alpha}{\pi} \left[\left(\frac{1}{2} - \ln \frac{m_V}{m_\ell} \right) \left(\ln r + \frac{1}{3} \right) + \frac{1}{2} \left(L_2(r) - \frac{\pi^2}{6} \right) + \frac{17}{72} + R(r) + \underbrace{\frac{r^2}{16} - \frac{3}{8}r}_{\text{missing from [10]}} \right]$$

$$r = \frac{2\omega_{cut}}{m_V}, \quad R(r) = \left(r - \frac{r^2}{4} - \frac{3}{4} \right) \left[\ln \left(\frac{m_V}{m_\ell} \sqrt{1-r} \right) - \frac{1}{2} \right], \quad L_2(x) = - \int_0^x \frac{\ln(1-t)}{t} dt$$

For muons add the electron-loop vacuum polarization = δ_{self} of Eq. (8) with m_e . Equivalently the missing piece is $+(\alpha/\pi)(3r/2 - r^2/4)$ added to δ ; it vanishes as $r \rightarrow 0$ and reaches $5\alpha/4\pi = 0.0029$ at $r = 1$.

Status of this formula. It is *not* exact in the lepton mass, and not the $m_\ell = 0$ limit either: the mass is kept in the logarithms, while terms of relative order $(m_\ell/E)^2$, $E = m_V/2$, are dropped. Ehlotzky & Mitter say so above their Eq. (22) (“neglecting terms of order $(m/E)^2$ ”). Numerically $(m_e/E)^2 = 1.1 \times 10^{-7}$ and $(m_\mu/E)^2 = 4.7 \times 10^{-3}$, so at $r_{max} = 1$ it is off by 1×10^{-7} for electrons and -7×10^{-5} for muons, against the exact Eq. (28).

2 The missing term is Ehlotzky’s error, not yours

Your Eqs. (14), (15), (19), (20) are *faithful transcriptions* of Ehlotzky & Mitter (E&M) Eqs. (24) and (26):

$$\begin{array}{ccccccc} \text{VPK (11)} \leftrightarrow \text{E\&M (22)} & \text{VPK (13)} \leftrightarrow \text{E\&M (23)} & \text{VPK (14)} \leftrightarrow \text{E\&M (24)} & \text{VPK (18)} \leftrightarrow \text{E\&M (25)} \\ & & \text{VPK (19)} \leftrightarrow \text{E\&M (26)} & \end{array}$$

E&M Eq. (23), the differential spectrum, is correct. The term was lost when that spectrum was integrated analytically in 1968. Their Eq. (26) and their Table I inherit it. Recomputing their Table I from their own Eq. (26) reproduces **every printed digit**, electrons and muons:

r_{max}	E&M δ	Eq. (26) as published	corrected
0.10	−0.087	−0.087	−0.087
0.25	−0.034	−0.034	−0.033
0.50	−0.002	−0.002	+0.000
1.00	+0.021	+0.021	+0.024

The error has stood for 57 years. Your paper is the first to correct it.

3 Table I of the NLO paper: $m_V = 1.5$ GeV is correct

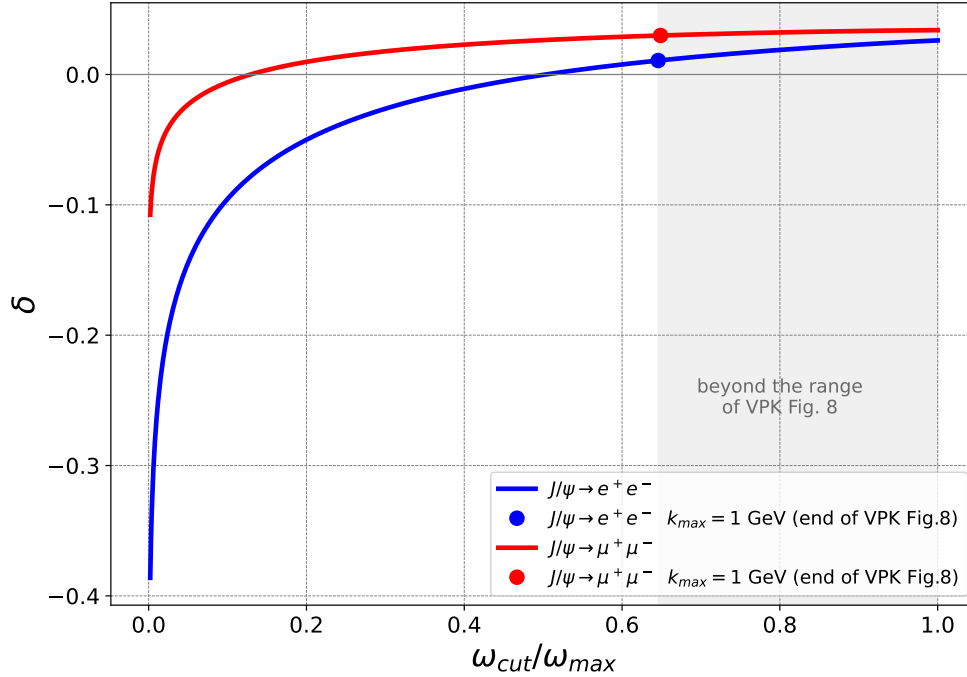
You suspected it was wrong; it is not. E&M's table column is $2E$ (MeV), and $2E = m_V$ because $E = m_V/2$ at resonance (stated below their Eq. 22). The rows are marked ρ at 765.0, ω at 782.0, ϕ at 1019.5 — meson masses. So the last row, 1500.0, is $m_V = 1.5$ GeV, exactly what the NLO paper used. δ_R settles it, depending on the mass only through $\ln(2E/m)$:

reading	δ_R	
$m_V = 1.5$ GeV (NLO's choice)	-0.746525	$\leftarrow$ E&M print -0.747
$m_V = 3.0$ GeV (if 1500 were E)	-0.868832	
$m_V = M(J/\psi)$	-0.874655	

What *is* wrong is their claim of agreement with that table: at $r_{max} = 1$ they have 0.023891 against E&M's 0.021. They agree only in the small- r_{max} rows.

4 NLO Fig. 4 vs VPK Fig. 8

Same quantity, different axes.



(a) **The endpoints differ.** NLO uses $\omega_{max} = m_V/2 - 2m_\ell^2/m_V$ (their Eq. 15); you use $k_{max} = M_V/2$. Identical for electrons, but for muons 1.54125 vs 1.54846 GeV, i.e. $r_{max}^{max} = 0.99535$, not 1 — a 0.47% shift.

(b) **Your Fig. 8 covers only the first two thirds, stopping at $k_{max} = 1.0$ GeV = $0.646\omega_{max}$.** Their Fig. 4 runs to 1.

$\omega_{cut}/\omega_{max}$	$k_{max}(e)$	$\delta(e)$	$k_{max}(\mu)$	$\delta(\mu)$
0.10	0.1548	-0.095444	0.1541	-0.005934
0.25	0.3871	-0.036636	0.3853	+0.014282
0.50	0.7742	-0.000275	0.7706	+0.026393
1.00	1.5485	+0.026135	1.5412	+0.034023

At the endpoint against their exact Bardin–Shumeiko numbers: e^+e^- yours 0.026135 vs exact 0.026135 (diff 10^{-7}); $\mu^+\mu^-$ yours 0.034017 vs exact 0.034097 (diff -8×10^{-5} , the expected URA limit).

5 New finding: your $k_{max} = M_V/2$ contradicts your own Eq. (22)

The text above your Eq. (16) says the maximum photon energy is $k_{max} = M_V/2$. That is the massless limit. The exact endpoint is

$$\omega_{max} = \frac{M_V^2 - 4m_\ell^2}{2M_V} = \frac{M_V}{2} - \frac{2m_\ell^2}{M_V},$$

which is NLO Eq. (15) — **and it is already in your paper**, as Eq. (22), $E_\gamma = (M_{J/\psi}^2 - M_{ee}^2)/(2M_{J/\psi})$ evaluated at $M_{ee} \rightarrow 2m_\ell$. So Eqs. (16) and (22) of your own paper disagree. No consequence for electrons. For muons r_{max} can only reach 0.99535, so the $r_{max} = 1$ row of a muon table lies outside phase space. **Not fixed — your call.**

6 Formula-by-formula status

VPK	NLO	verdict
(1)–(5)	(3)–(5)	agree; conventions differ (fixed λ vs polarization-averaged)
(6)	(1)+(2)	agree exactly. Fixed today: $4m_\ell^2/m_V \rightarrow 4m_\ell^2/m_V^2$
(7)	(6)	identical (sympy: difference 0)
(8)	—	ultrarelativistic limit; NLO has none
(11),(12)	(8)+(12)+(13)	identical to machine precision
(13)	(14) integrand	identical
(14),(15)	—	were missing the term. Fixed today
(16)	(15)	disagree — see Section 5
(19),(20)	—	were missing the term. Fixed today; Eq. (20) $m_e \rightarrow m_\mu$ also fixed
(21),(22)	—	no counterpart; (22) is equivalent to NLO (15)
—	(16)	FSIC $3\alpha/4\pi$; your corrected Eq. (19) reproduces it
—	(28)	exact Bardin–Shumeiko; no counterpart in your paper

7 Changed today

Paper (Jpsi_RC.tex, compiled and published): Eqs. (14),(15),(19),(20) missing term added; Eq. (20) leading log $m_e \rightarrow m_\mu$; Eq. (6) $4m_\ell^2/m_V \rightarrow 4m_\ell^2/m_V^2$; Fig. 7 recomputed at $\varepsilon = 0.001$ (was 0.01, caption said 0.001); Figs. 3–8 in-plot titles removed, information moved to captions; Figs. 3,4,6 captions now name the two decay modes.

Code: three duplicate copies collapsed into one library `rc_vector_mesons.py`; `validate_rc.py` checks it against an independent quadrature reference and two analytic anchors — all pass. Repo `~/jpsi-rc-2026`, <https://github.com/vkubarovsky/jpsi-rc-2026>, mirrored to OneDrive Work/2026_Jpsi_rad

8 Waiting on you

- `notes/NLO_comments.pdf` — the note for your co-authors: all six of your items plus the Eq. (4) polarization-sum point. *Not sent anywhere.*
- Section 5: Eq. (16) vs Eq. (22) inconsistency — not fixed.
- “Formala” $\times 2$ in the Fig. 9 legend — you said leave Fig. 9 alone.
- `fig_igor` defined twice in `Jpsi_rad_corr.py`.
- Never verified: your Eqs. (2)–(5) in detail, (17), (18); and the Bethe–Heitler formulas behind the -20% claim in Section VII.